Climate Change

WHAT ARE THE KEY ISSUES THAT SHAPE OUR SUSTAINABLE FUTURE?

The world has to reduce greenhouse gas emissions to net zero by 2055 to avoid a devastating outcome.

Since the Paris Agreement on climate change was officially signed in 2016, further evidence has shown that the global average temperature needs to stay below 1.5°C above pre-industrial levels to avoid a devastating outcome. To achieve this, the world must reduce greenhouse gas (GHG) emissions to net zero by 2055. Considering that current global efforts are still very far from sufficient to achieve this goal, every stakeholder, including us, is expected to set more ambitious targets to decarbonize and contribute to this common goal.

The frequency and magnitude of extreme weather events are expected to increase.

It is clear that the frequency and magnitude of extreme weather events are increasing, not least in Hong Kong, which was hit by a super typhoon for two consecutive years, resulting in substantial losses. The same phenomenon is happening in mainland China, where the precipitation pattern is much harder to predict than before. For a property developer and building operator, extreme weather events can potentially result in asset damages, business interruption and even loss of lives. We must, therefore, build up our climate resilience and adaptive capacity to cope with the changing climate.

CLIMATE CHANGE

Our 2030 Ambition

We aim to reduce at least

- 50% of our overall carbon intensity against our 2015 baseline
- 10% of our absolute carbon emission against our 2017 baseline

Relevant GRI Disclosures: 302 (2016), 303 (2018), 305 (2016), 307 (2016)

Policies are evolving fast to facilitate the transition to a low-carbon economy.

While decarbonization is the ultimate goal, governments need to adopt a multi-pronged approach, such as changing the fuel mix for electricity generation, regulating building energy consumption and promoting green transportation, in order to realize this goal. It is anticipated that all policies relevant to the transition to a low-carbon economy will evolve fast. We must act swiftly and collaborate with governments and the business sector in order to stay ahead of the policies and remain agile to change.

Water scarcity remains a neglected issue that is intensifying at an alarming speed.

Although water scarcity has been globally recognized, it is still woefully neglected by businesses, as the adverse impact of water scarcity is hardly felt due to the advanced water infrastructure in cities. With water availability expected to decrease further as a result of climate change, we need to act immediately before water scarcity becomes a critical and irreversible challenge that could interrupt our business operations in the future.

MAJOR DEVELOPMENTS IN 2018

Enhancing Climate Governance and Management of Climate Risks

In 2018, the Group officially included climate risks as one of the corporate risks under our Enterprise Risk Management framework, which is reported to the Audit Committee and the Board on a quarterly basis. This ensures Board oversight of the climate risks relevant to our business.

By the end of 2018, we had commenced a preliminary climate risk mapping exercise at the Group level, in which we outlined the physical and transitional risks of climate change that are relevant to our business, as well as the potential consequences of those risks. In 2019, we shall proceed to conduct an in-depth climate impact assessment for a pilot site so that we can acquire sufficient information to facilitate risk prioritization and resilience planning at site level.

Building Climate Resilience

The Group continued to build up the resilience of our business to growing climate risks. We reviewed and revamped the Business Continuity Plans of our projects across the Group in 2018, which cover scenarios of extreme weather events that may potentially halt our business operations and outline how our teams should react to minimize the adverse impacts. Conscious that flooding has caused significant interruption to our business in the past few years, notably the severe flood that hit Riverside 66 in Tianjin in 2016, we have taken various measures to build our resilience during the past two years. These include the installation of flood gates, increasing stocks of sand bags and water pumps, and capacity building for staff, so as to reinforce the flood prevention capability of our projects on the Mainland.

Accounting for Our Greenhouse Gas Emissions

To keep track of and manage our impact on climate change, we regularly monitor and review our greenhouse gas emissions. In addition, we have appointed an independent consultant to verify our direct GHG emissions (Scope 1) and energy indirect GHG emissions (Scope 2) of our building operations, aiming to ensure that our carbon footprint data are accurate and reliable. Currently, our building energy consumption accounts for over 99% of our Scope 1 and Scope 2 emissions.

Selected indirect GHG emissions (Scope 3) incurred by our direct operations have also been calculated and reported to demonstrate our commitment to transparency. However, it remains beyond our capacity to estimate the GHG emissions of our supply chain.

WHAT'S NEXT?

We will start to develop a climate impact assessment framework at site level in 2019 to facilitate regular assessments in the forthcoming years.

Reinforcing Our Decarbonization Strategy

To reduce our buildings' energy consumption and decarbonize our business, we have adopted five major approaches as outlined below:

DESIGN AND BUILD ENERGY EFFICIENT AND ENVIRONMENTALLY-FRIENDLY BUILDINGS	We design and build our buildings that achieve at least Gold rating under the LEED Certification Program. This ensures that our buildings are energy efficient with low environmental footprint based on international standard. Please refer to the "Green Building" section of this Report for more details.
OPTIMIZE ENERGY PERFORMANCE OF EXISTING BUILDINGS	We optimize the energy performance of our existing buildings from time to time through our Asset Enhancement Initiative, under which we retrofit and replace aged and inefficient building facilities, as well as optimize building facilities through automation and retro-commissioning. For instance, we continued to implement various energy saving initiatives in our old properties in Hong Kong, such as the installation of LED lightings and variable speed drive for a cooling tower's fan motor. It is calculated that such efforts have resulted in a total annual energy saving of up to 58,300 kWh.
LOWER RESOURCES CONSUMPTION THROUGH BETTER OPERATIONS MANAGEMENT	Through better operations management, e.g. switching off unnecessary external lightings, lifts and escalators at night, we strive to reduce electricity consumption of our properties. Besides, we have avoided business activities that generate extra carbon emissions. For instance, we reduced the frequency of group-level management meetings in 2018. It is estimated that around 400 round-trips of flight can be avoided annually as a result of this new arrangement.
REDUCE RELIANCE ON FOSSIL-FUEL BASED ELECTRICITY	To reduce reliance on fossil-fuel based electricity, we strive to utilize renewable energy at our properties wherever feasible. We have installed solar panels at Palace 66, Parc 66, Center 66, Forum 66, Riverside 66 and Olympia 66 in mainland China to generate electricity. In 2018, 756,380 kWh of our electricity consumption was generated from solar panels.
MANAGE GREENHOUSE GAS EMISSIONS IN THE SUPPLY CHAIN	While we find it challenging to proactively measure and manage the GHG emissions in our supply chain, we are committed to working with our suppliers to reduce the environmental footprint of our value chain through policies and communications. Please refer to the "Sustainable Procurement" section of this Report for more details.

In terms of a year-to-year comparison, despite our efforts at energy conservation, we recorded only a 4.6% reduction for absolute greenhouse gas (GHG) emissions (Scope 1 & 2) and a 3% reduction on the carbon intensity of our Hong Kong portfolio, due to the higher average temperature in 2018. On the other hand, we saw a similar situation with our Mainland portfolio, where the GHG emissions intensity (Scope 1 & 2) increased by 10.3%, while absolute GHG emissions (Scope 1 & 2) increased by 21.9%, due to the expansion of our reporting scope to cover Olympia 66 in Dalian in 2018.

For our electricity-incurred GHG emission intensity, we recorded a 9.4% reduction, compared to our 2015 baseline, 5.6% away from the 15% interim reduction target by 2021.

Performance Tables

Economic

INDICATOR	UNIT	2016			2017			2018		
		HK	MC	Total	HK	MC	Total	HK	MC	Total
ECONOMIC VALUE GENERATED										
Revenue (including property sales revenue)	HK$ million	9,064	3,995	13,059	7,241	3,958	11,199	**5,164**	**4,244**	**9,408**
ECONOMIC VALUE DISTRIBUTED										
Operating cost	HK$ million	2,659	1,481	4,140	1,785	1,504	3,289	**1,081**	**1,505**	**2,586**
Employee wages and benefits		906	468	1,374	820	607	1,427	**873**	**695**	**1,568**
Borrowing cost capitalization		N/A	N/A	223	N/A	N/A	56	**N/A**	**N/A**	**251**
Interest and other borrowing costs paid		N/A	N/A	1,287	N/A	N/A	1,157	**N/A**	**N/A**	**1,245**
Dividends paid		N/A	N/A	3,373	N/A	N/A	3,373	**N/A**	**N/A**	**3,374**
Payments to government (all taxes and related penalties)		850	602	1,452	716	588	1,304	**489**	**635**	**1,124**
Community investments		N/A	N/A	19	N/A	N/A	16	**N/A**	**N/A**	**15**
Number of cities of operation	Number	1	8	9	1	8	9	**1**	**9**	**10**
ECONOMIC VALUE RETAINED										
Economic value retained	HK$ million	N/A	N/A	2,741	N/A	N/A	1,959	**N/A**	**N/A**	**989**

HK: Hong Kong
MC: Mainland China

Environmental

INDICATOR	UNIT	2016			2017			2018		
		HK	MC	Total	HK	MC	Total	HK	MC	Total
ENERGY CONSUMPTION AND GENERATION [(1),(2),(3),(4),(5),(6)]										
Direct energy consumed by type										
Petrol by vehicles	GJ	N/A	N/A	N/A	402	N/A	N/A	**402**	**N/A**	**N/A**
Diesel by vehicles and vessels	GJ	N/A	N/A	N/A	440	N/A	N/A	**168**	**N/A**	**N/A**
Diesel by emergency generators	GJ	88	249	336	56	195	251	**87**	**141**	**228**
Natural gas	GJ	0	4,448	4,448	0	4,278	4,278	**0**	**4,075**	**4,075**
Indirect energy consumed (all non-renewable by type)										
Electricity	kWh	78,256,239	151,801,871	230,058,110	73,113,501	167,733,995	240,847,495	**74,101,311**	**194,448,350**	**268,549,661**
Electricity	GJ	281,722	546,487	828,209	263,209	603,842	867,051	**266,765**	**700,014**	**966,779**
Electricity intensity of buildings in use	kWh/m²/year (CFA)	106.08	82.52	N/A	99.11	79.37	N/A	**101.97**	**79.09**	**N/A**
Hot water	GJ	N/A	N/A	N/A	0	143,510	143,510	**0**	**245,606**	**245,606**
Steam	GJ	N/A	N/A	N/A	0	39,274	39,274	**0**	**37,566**	**37,566**
Energy generated and consumed										
Renewable energy	kWh	156	501,916	502,072	26	521,405	521,431	**0**	**756,380**	**756,380**
GREENHOUSE GAS (GHG) EMISSIONS [(1),(2),(3),(7),(8)]										
Direct emissions (Scope 1)										
Building operation	Tonnes CO_2e	1,210.48	1,074.83	2,285.31	2,628.84	2,339.07	4,967.91	**1,204.37**	**1,858.28**	**3,062.65**
Company vehicles and vessels [(9)]	Tonnes CO_2e	N/A	N/A	N/A	61.81[(10)]	N/A	N/A	**41.45**	**N/A**	**N/A**
Energy indirect emissions (Scope 2)										
Building operation	Tonnes CO_2e	49,985.31	159,049.12	209,034.43	48,897.65	182,466.17	231,363.82	**47,996.88**	**223,242.65**	**271,239.53**
Other indirect emissions (Scope 3)										
Business air travel	Tonnes CO_2e	343.17	106.83	449.99	490.29	249.54	739.83	**586.63**	**251.51**	**838.22**
Emission Intensity										
Building operation [(11)]	Tonnes CO_2e/m²/year (CFA)	0.0694	0.0870	N/A	0.0698	0.0874	N/A	**0.0677**	**0.0964**	**N/A**

INDICATOR	UNIT	2016 HK	2016 MC	2016 Total	2017 HK	2017 MC	2017 Total	2018 HK	2018 MC	2018 Total
MATERIAL CONSUMPTION [(1),(2),(3)]										
Office paper	kg	N/A	N/A	N/A	N/A	N/A	N/A	**27,472**	**N/A**	**N/A**
WASTE MANAGEMENT [(1),(2), (3)]										
Waste disposal										
Municipal solid waste	Tonnes	5,608	16,476	22,084	7,864[(12)]	20,258	28,122	**7,735**	**22,063**	**29,798**
Construction waste by tenants [(13)]	Tonnes	N/A	50,354	N/A	N/A	157,970	N/A	**N/A**	**109,982**	**N/A**
Hazardous waste (Fluorescent light bulbs/tubes)	kg	10,628	1,181	11,809	5,176	1,776	6,952	**1,355[(14)]**	**1,511**	**2,866**
Recycled waste										
Paper	kg	110,837	509,400	619,980	102,028	389,957	491,985	**84,277**	**392,047**	**476,324**
Metal	kg	1,387	55,440	56,792	1,810	17,945	19,755	**88**	**32,919**	**33,007**
Food waste	kg	55,211	N/A	55,211	15,907	N/A	15,907	**89,145**	**N/A**	**89,145**
Plastics	kg	2,239	50,417	52,626	1,724	50,316	52,040	**1,259**	**51,637**	**52,896**
Glass	kg	4,727	57,968	61,722	5,242	37,596	42,838	**4,123**	**30,226**	**34,349**
WATER MANAGEMENT [(1),(2), (3), (15)]										
Fresh water consumption	ML	508.609	1,367.790	1,876.399	554.898	1,207.754	1,762.652	**526.784**	**1,379.484**	**1,906.268**
Water intensity of buildings in use	m^3/m^2/year	1.42	1.19	N/A	1.55	0.99	N/A	**1.50**	**0.94**	**N/A**

GRI Standards Disclosure		HKEx ESG Guide General Disclosure (GD) and KPIs	Description	Section/ Explanation	Page
TOPIC-SPECIFIC DISCLOSURES					
Economic					
Economic Performance (2016)	103-1 103-2 103-3	-	Management approach disclosure	About the Group Sustainability in Action	3-4 6-15
	201-1		Direct economic value generated and distributed	Performance Tables - Economic	52
	201-2		Financial implications and other risks and opportunities due to climate change	Future of Space	27-37
	201-3		Defined benefit plan obligations and other retirement plans	We report on our benefit plan obligations and retirement benefits for employees in the 2018 Annual Report (P.196-200).	N/A
Market Presence* (2016)	103-1 103-2 103-3		Management approach disclosure	About the Group	3-4
	202-2		Proportion of senior management hired from the local community at significant locations of operation	93% of our senior management (Director grade and above) was hired from the local community, i.e. Hong Kong and the Mainland. The profiles of the directors and key executives are outlined in the 2018 Annual Report (P.125-133).	N/A
Indirect Economic Impacts* (2016)	103-1 103-2 103-3		Management approach disclosure	Sustainability in Action	6-15
	203-1		Infrastructure investments and services supported	Future of Space	27-37
	203-2		Significant indirect economic impact		
Procurement Practices* (2016)	103-1 103-2 103-3	Aspect B5 GD KPI B5.2	Management approach disclosure	Best-in-Class Operating Practices	38-48
	204-1	-	Proportion of spending on local suppliers	We only disclose the aggregate procurement spending in Hong Kong. Our total procurement spending for Hong Kong operations was HK$146,828,737, all of which was spent on local suppliers.	N/A
	-	KPI B5.1	Number of suppliers by geographical region	We purchased goods and services from 287 suppliers for our Hong Kong operations in the reporting period. It is currently beyond our capacity to estimate the number of suppliers for mainland China.	N/A
Anti-corruption (2016)	103-1 103-2 103-3	Aspect B7 GD KPI B7.2	Management approach disclosure	Sustainability in Action Best-in-Class Operating Practices 2018 Annual Report (P.120-121)	6-15 38-48 N/A
	205-2	-	Communication and training about anti-corruption policies and procedures	Best-in-Class Operating Practices Performance Tables – Social	38-48 55-59
	205-3	KPI B7.1	Confirmed incidents of corruption and actions taken	There were no confirmed incidents of corruption during the reporting period.	N/A

GRI Standards Disclosure		HKEx ESG Guide General Disclosure (GD) and KPIs	Description	Section/ Explanation	Page
Environmental					
Material*	-	KPI A2.5	Total packaging material used for finished products and if applicable, with reference to per unit produced	This KPI is not applicable to our business.	N/A
Energy (2016)	103-1 103-2 103-3	Aspect A2 GD Aspect A3 GD KPI A2.3 KPI A3.1	Management approach disclosure	Sustainability in Action Future of Space Environmental Policy	6-15 27-37 N/A
	302-1	KPI A2.1	Energy consumption within the organization	Performance Tables – Environmental	53-54
	302-3	KPI A2.1	Energy intensity		
	302-4	KPI A2.3	Reduction of energy consumption	Future of Space	27-37
Water* (2018)	103-1 103-2 103-3	Aspect A2 GD Aspect A3 GD KPI A2.4 KPI A3.1	Management approach disclosure	Future of Space During the reporting period, we did not encounter any problems in sourcing water for our operations. Environmental Policy	27-37 N/A N/A
	303-5	KPI A2.2	Water consumption	Performance Tables – Environmental	53-54
	-	KPI A2.2	Water intensity		
Emissions* (2016)	103-1 103-2 103-3	Aspect A1 GD Aspect A3 GD KPI A1.5 KPI A3.1	Management approach disclosure	Future of Space Environmental Policy	27-37 N/A
	305-1	KPI A1.1 KPI A1.2	Direct (Scope 1) GHG emissions	Performance Tables – Environmental	53-54
	305-2	KPI A1.1 KPI A1.2	Energy indirect (Scope 2) GHG emissions		
	305-3	KPI A1.1 KPI A1.2	Other indirect (Scope 3) GHG emissions		
	305-4	KPI A1.2	GHG emissions intensity		
Effluents and Waste* (2016)	103-1 103-2 103-3	Aspect A1 GD KPI A1.5 KPI A1.6 KPI A3.1	Management approach disclosure	Future of Space	27-37
	306-2	KPI A1.3 KPI A1.4 KPI A1.6	Weight of waste by type and disposal method	Performance Tables – Environmental	53-54
Environmental Compliance (2016)	103-1 103-2 103-3	Aspect A1 GD	Management approach disclosure	Sustainability in Action Future of Space 2018 Annual Report (P.116-120)	6-15 27-37 N/A
	307-1		Non-compliance with environmental laws and regulations	We were not issued with any significant fine or sanctions for any non-compliance with environmental laws and regulations.	N/A

GRI Standards Disclosure		HKEx ESG Guide General Disclosure (GD) and KPIs	Description	Section/ Explanation	Page
Social					
Employment (2016)	103-1 103-2 103-3	Aspect B1 GD	Management approach disclosure	Sustainability in Action Future of Experience	6-15 17-26
	401-1	KPI B1.2	New employees hiring and employee turnover	Performance Tables – Social	55-59
	401-2	Aspect B1 GD	Benefits provided to full-time employees that are not provided to temporary or part-time employees	Future of Experience 2018 Annual Report (P.196-200)	17-26 N/A
	401-3	-	Parental leave	Performance Tables – Social	55-59
Occupational Health and Safety (2018)	103-1 103-2 103-3	Aspect B2 GD KPI B2.3	Management approach disclosure	Sustainability in Action Future of Experience Best-in-Class Operating Practices	6-15 17-26 38-48
	403-9	KPI B2.1 KPI B2.2	Number, rates and types of work-related injuries, lost days due to work-related injuries and, number and rate of work-related fatalities	Construction site workers: Best-in-Class Operating Practices Employees: Future of Experience Performance Tables – Social Other supervised workers: No injury or accident has been reported by our contractors	38-48 17-26 55-59 N/A
Training and Education (2016)	103-1 103-2 103-3	Aspect B3 GD	Management approach disclosure	Sustainability in Action Future of Experience	6-15 17-26
	404-1	KPI B3.2	Average hours of training per year per employee	Performance Tables – Social	55-59
	-	Aspect B3 GD	Programs for upgrading employee skills and transition assistance programs	Future of Experience Performance Tables – Social	17-26 55-59
	404-3	-	Percentage of employees receiving regular performance and career development reviews	Performance Tables – Social	55-59
	-	KPI B3.1	The percentage of employees trained by gender and employee category	All employees received training in the reporting period.	N/A
Diversity and Equal Opportunity* (2016)	103-1 103-2 103-3	Aspect B1 GD	Management approach disclosure	Future of Experience	17-26
	405-1	-	Diversity of governance bodies and employees	Performance Tables – Social 2018 Annual Report (P.108)	55-59 N/A
Non-discrimination* (2016)	103-1 103-2 103-3	Aspect B1 GD	Management approach disclosure	Future of Experience	17-26
	406-1	-	Incidents of discrimination and corrective actions taken	There were no confirmed incidents of discrimination in the reporting period.	N/A
Local Communities* (2016)	103-1 103-2 103-3	Aspect B8 GD	Management approach disclosure	Sustainability in Action Best-in-Class Operating Practices	6-15 38-48
	413-2	-	Operations with significant actual and potential negative impacts on local communities	No significant actual or potential negative impacts were identified during the reporting period.	N/A
	-	KPI B8.1	Focus areas of contribution	Best-in-Class Operating Practices	38-48
		KPI B8.2	Resources contributed to the focus area		